

# From Coccyx Pain to Neurological Impairment: Investigating a Case of Pachymeningeal Enhancement

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## Introduction

Cryptococcal meningitis and cytomegalovirus (CMV) chorioretinitis are well-recognized opportunistic infections in patients with AIDS; however, concurrent presentation poses significant diagnostic and therapeutic challenges. This case highlights the importance of multidisciplinary collaboration in addressing both acute infectious concerns and long-term healthcare access for a critically ill, immunocompromised patient.

## Case Presentation

A 35-year-old woman with a PMHx of PCOS and HLD presented with a one-month history of progressive coccyx pain and constipation, initially attributed to coccydynia following a vaginal delivery three months prior. She eventually returned presenting with severe sepsis and significant bilateral lower extremity weakness and a new inability to ambulate. A pan-CT revealed no evidence of infection or malignancy. However, MRI revealed diffuse pachymeningeal and leptomeningeal enhancement extending from T8 to the sacrum. A lumbar puncture was attempted but yielded only 2 mL of cerebrospinal fluid (CSF) due to diffuse leptomeningeal thickening. During the hospital course, the patient developed tender, erythematous nodules on her lower extremities, consistent with erythema nodosum.

## Diagnostic Challenges

- **Neurosarcoidosis** is a rare manifestation of sarcoidosis, characterized by **granulomatous** inflammation of the nervous system, commonly causing **cranial neuropathies, pachymeningitis, and hydrocephalus**
- **IgG4-RD** is a systemic immune-mediated condition characterized by fibroinflammatory processes that **can affect nearly any organ**, typically involving the pancreas, bile ducts, salivary glands, kidneys, or retroperitoneal structures
- While the **normal serum ACE level and negative chest imaging** do not support pulmonary sarcoidosis, these findings **do not exclude isolated neurosarcoidosis**, as ACE sensitivity is variable and non-pulmonary involvement is common
- **Serum IgG4 is elevated in approximately two-thirds of cases**, though it is neither sensitive nor specific on its own
- A **leptomeningeal biopsy is the gold standard** for diagnosing both neurosarcoidosis (via confirmation of non-caseating granulomas) and IgG4-related disease (via identification of IgG4-positive plasma cells, storiform fibrosis, and obliterative phlebitis)
- Both neurosarcoidosis and IgG4-RD are steroid-responsive conditions, but **long-term management should focus on steroid-sparing agents like mycophenolate or methotrexate**

## Discussion

In the absence of sufficient CSF analysis, the utility of emerging biomarkers becomes increasingly relevant. Emerging biomarkers for neurosarcoidosis include soluble IL-2 receptor (sIL-2R), which correlates with disease activity and can guide diagnosis when sarcoidosis is suspected. For IgG4-RD, elevated IgG4:IgG ratio and serum eosinophil counts are being explored as potential markers to support diagnosis in challenging cases.



**Figure 1** T2-weighted sagittal MRI of the cervical, thoracic, and lumbar spine (from left to right)

**Table 1:** Systematic evaluation of pachymeningitis using the MIND-V framework

| MIND-V Category                        | Findings in This Case                                                                                                                                              | Conclusion                                                             |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>M</b> — Malignancy                  | Pan-CT negative for neoplasm; no evidence of malignancy on MRI.                                                                                                    | Malignancy unlikely                                                    |
| <b>I</b> — Infection                   | CSF viral panel, CSF culture, TB, and Histo all negative; lymphocytic pleocytosis                                                                                  | Infection unlikely                                                     |
| <b>N</b> — Non-infectious Inflammatory | Elevated ANA (1:1280) with elevated ESR and CRP<br>RF/Anti-CCP, ANCA (PR3/MPO), Anti-dsDNA, Anti-Sm, SSA/SSB Negative<br>Serum IgG4 — Normal<br>Serum ACE — Normal | Autoimmune process suspected; narrowed to IgG4-RD and neurosarcoidosis |
| <b>D</b> — Drug-Related                | No history of intrathecal drugs or other medications known to cause pachymeningitis.                                                                               | Drug-induced unlikely                                                  |
| <b>V</b> — Vascular                    | Normal opening pressure and normal closing pressure on LP; no signs of low ICP.                                                                                    | Low ICP unlikely                                                       |

